3. Residential Projects Catalog: Family Houses & Timber Structures

Document ID: DOC-03-RES | **Version:** 3.4 (2026) | **Division:** Residential Development
Primary OWL Classes: `ResidentialProject`, `FamilyHouse`, `FamilyHouseExample`, `WoodenHouse`, `ModernProject`
Primary OWL Properties: `usesMaterial`, `usesMachine`, `hasEmployee`, `hasDesigner`, `hasSiteManager`, `year`

3.1 Ontological Definition of Residential Projects

In the ApexBuild ontology, `ResidentialProject` is a direct subclass of `BuildingProject` (which in turn is a subclass of `Project`). Consequently, every `ResidentialProject` inherits all structural requirements of a `BuildingProject`:

- `usesMaterial some Material` (explicitly stated on both `BuildingProject` and `ResidentialProject`).
- `usesMachine some ConstructionMachine` (inherited from `BuildingProject` and `Project`).
- `hasCustomer some Customer` and `hasSiteManager some SiteManager` (inherited from `BuildingProject`).
- Connections to staff: Every `Designer` has restriction `hasProject some ResidentialProject`, and every `Plumber` has restriction `worksOn some ResidentialProject`.

The class `ResidentialProject` branches into two main housing categories: `FamilyHouse` (single-family detached or semi-detached residences) and `ApartmentBuilding` (multi-unit residential complexes, covered in Document 04).

3.2 Standard Family House (Class: FamilyHouse)

A standard `FamilyHouse` in our ontology represents modern suburban residences built during the 21st century. Formally, `FamilyHouse` is a subclass of `ResidentialProject` with three class restrictions:

1. **Chronological Restriction:** `year some integer[≥ 2000, < 2026]` for standard historical catalog models (automatically making every such instance a `ModernProject`, since `ModernProject ≡ year ≥ 2000`).
2. **Machinery Requirement:** `usesMachine some ConstructionMachine`.
3. **Material Requirement:** `usesMaterial some Material`.

3.3 Detailed Specification: FamilyHouseExample (2026 Flagship Villa)

The ontology includes a dedicated flagship class `FamilyHouseExample` (subclass of `FamilyHouse`) representing our 2026 Model Villa "Vila Záhorie" constructed in Záhorská Bystrica for client Dr. Tomáš Gregor (CUST-103). This specific project is defined by five explicit OWL restrictions in `stavebna_firma_ontology – kópia.owl`:

> **Formal OWL Restrictions for `FamilyHouseExample`:**
>
> - `rdfs:subClassOf FamilyHouse`

- `usesMaterial some Concrete` (C25/30 waterproof foundation slab and ceiling monoliths)
- `usesMaterial some Brick` (Porotherm 38 Ti Profi precision ceramic thermal blocks)
- `usesMachine some Excavator` (CAT 320 Hydraulic Crawler Excavator for foundation pit & utility trenches)
- `hasEmployee some ConstructionWorker` (On-site masonry and concrete crew led by Štefan Baláž)
- `year value 2026` (`owl:hasValue 2026`)

Parameter	Specification for `FamilyHouseExample` ("Vila Záhorie 2026")
Project Code & Completion Year	PRJ-RES-2026-01 | `year = 2026` (classified as `ModernProject`)
Floor Area & Storeys	245 m² usable floor area | 2 storeys + underground technical basement
Primary Structural Materials	`Concrete` (110 m³ of C25/30 XC2 for foundations & slabs) and `Brick` (4,800 pcs ceramic blocks)
Assigned Machinery	`Excavator` (CAT 320, Fleet ID: EXC-01) for 420 m³ earthworks
Assigned Personnel	`SiteManager`: Ing. Martin Kováč | `Designer`: Ing. Elena Varga | `ConstructionWorker`: Štefan Baláž | `Plumber`: Milan Krátky
Total Turnkey Budget	EUR 485,000 (excl. VAT)

3.4 Sustainable Timber Housing: Class WoodenHouse

For ecological construction, the ontology defines `WoodenHouse` as a specialized subclass of `FamilyHouse`. Unlike conventional masonry houses, `WoodenHouse` carries a universal (`owl:allValuesFrom`) material restriction:

> **Formal OWL Axiom for `WoodenHouse`:**
> `WoodenHouse` ⊑ `FamilyHouse` and `usesMaterial only Wood` (`owl:allValuesFrom Wood`).
> *Important Ontological Consequence:* Because `Concrete` is declared explicitly disjoint with `Wood` (`Concrete owl:disjointWith Wood`), a pure `WoodenHouse` instance in the ontology cannot have a `usesMaterial` assertion pointing to an instance of `Concrete` without triggering a logical inconsistency in an OWL DL reasoner!

In practice, our **Tatry CLT WoodenHouse Cluster (PRJ-RES-2025-04)** utilizes Cross-Laminated Timber (CLT) spruce panels (class `Wood`, strength grade C24 / GL24h) mounted on screw-pile timber substructures, achieving passive-house thermal performance with a construction time of just 14 weeks per unit and a unit cost of EUR 310,000.